EXPERIMENT 14 – File Allocation Strategies

Aim: To implement Sequential Allocation, Indexed Allocation, and Linked Allocation.

Program A – Sequential File Allocation

C Program:

```
#include <stdio.h>

int main()
{
    int start, length, i;

    printf("Enter Starting Block: "); scanf("%d", &start);
    printf("Enter File Length (Number of Blocks): "); scanf("%d", &length);

    printf("\nAllocated Blocks:\n");
    for(i = 0; i < length; i++)
        printf("%d ", start + i);

    printf("\n");
    return 0;
}
```

Output (Starting Block=10, Length=5):

```
Allocated Blocks:
10 11 12 13 14
```

Program B – Indexed File Allocation

Shell Script:

```
#!/bin/bash

echo "Enter Index Block:"
read index

echo "Enter Number of Blocks:"
read n

echo "Enter Block Numbers:"
for ((i=0;i<n;i++))
do
    read block[$i]
done

echo "Index Block : $index"
echo -n "Allocated Blocks : "
for ((i=0;i<n;i++))
do
    echo -n "${block[$i]} "
done
echo
```

Program C – Linked File Allocation

C Program:

```
#include <stdio.h>

int main()
{
    int n, blocks[20], i;

    printf("Enter Number of Blocks: "); scanf("%d", &n);
    printf("Enter Block Numbers:\n");
    for(i = 0; i < n; i++) scanf("%d", &blocks[i]);

    printf("\nLinked Allocation:\n");
    for(i = 0; i < n - 1; i++)
        printf("%d --> ", blocks[i]);

    printf("%d --> NULL\n", blocks[n - 1]);

    return 0;
}
```

Output (Blocks: 10 20 35 42 57):

```
Linked Allocation:
10 --> 20 --> 35 --> 42 --> 57 --> NULL
```

Result: Sequential, Indexed, and Linked File Allocation strategies implemented and verified.
